Matrix Theory I: Matrices, Trace and Determinants

Problem sheet for the Day 1 class · CSIR NET previous year questions

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. CSIR NET June 2020 MCQ Let A be a 2×2 real matrix with $\det A = 1$ and $\operatorname{tr} A = 3$. The value of $\operatorname{tr}(A^2)$ is

(A) 2
(B) 10
(C) 9
(D) 7

2. CSIR NET Dec 2016 MCQ Let $D_1 = \det \begin{pmatrix} a & b & c \\ x & y & z \\ p & q & r \end{pmatrix}$ and $D_2 = \det \begin{pmatrix} -x & a & -p \\ y & -b & q \\ z & -c & r \end{pmatrix}$. Then

(A) $D_1 = D_2$
(B) $D_1 = 2D_2$
(C) $D_1 = -D_2$
(D) $2D_1 = D_2$

3. CSIR NET Dec 2016 MCQ Consider the matrix $A = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$, where $\theta = \frac{2\pi}{31}$. Then A^{2015} equals

(A) A
(B) I
(C) $\begin{pmatrix} \cos 13\theta & \sin 13\theta \\ -\sin 13\theta & \cos 13\theta \end{pmatrix}$
(D) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

4. CSIR NET Dec 2015 MCQ Let $A \neq I_n$ be an $n \times n$ matrix such that $A^2 = A$, where I_n is the identity matrix of order n. Which of the following statements is **false**?

(A) $(I_n - A)^2 = I_n - A$
(B) $\operatorname{Trace}(A) = \operatorname{Rank}(A)$
(C) $\operatorname{Rank}(A) + \operatorname{Rank}(I_n - A) = n$
(D) The eigenvalues of A are each equal to 1

5. CSIR NET June 2018 MCQ If A is a 2×2 matrix over $\mathbb{R}$ with $\det(A + I) = 1 + \det(A)$, then we can conclude that

(A) $\det(A) = 0$
(B) $A = 0$
(C) $\operatorname{Tr}(A) = 0$
(D) A is nonsingular

6. CSIR NET Dec 2014 MCQ Let A, B be $n \times n$ matrices such that $BA + B^2 = I - BA^2$, where I is the $n \times n$ identity matrix. Which of the following is always true?

(A) A is nonsingular
(B) B is nonsingular
(C) $A + B$ is nonsingular
(D) AB is nonsingular

7. CSIR NET Dec 2014 MCQ The determinant of the $n \times n$ permutation matrix with 1 on the anti-diagonal (entries $(i, n+1-i)$) and 0 elsewhere is ($\lfloor x \rfloor$ denotes the greatest integer not exceeding x)

(A) $(-1)^n$
(B) $(-1)^{\lfloor n/2 \rfloor}$
(C) -1
(D) 1

8. CSIR NET June 2016 MCQ Let A, B be $n \times n$ real matrices such that $\det A > 0$ and $\det B < 0$. For $0 \leq t \leq 1$, consider $C(t) = tA + (1-t)B$. Then

(A) $C(t)$ is invertible for each $t \in [0, 1]$
(B) there is a $t_0 \in (0, 1)$ such that $C(t_0)$ is not invertible
(C) $C(t)$ is not invertible for each $t \in [0, 1]$
(D) $C(t)$ is invertible for only finitely many $t \in [0, 1]$

9. CSIR NET Dec 2016 MSQ If P and Q are invertible matrices such that $PQ = -QP$, then we can conclude that

(A) $\mathrm{Tr}(P) = \mathrm{Tr}(Q) = 0$
(B) $\mathrm{Tr}(P) = \mathrm{Tr}(Q) = 1$
(C) $\mathrm{Tr}(P) = -\mathrm{Tr}(Q)$
(D) $\mathrm{Tr}(P) \neq \mathrm{Tr}(Q)$

10. CSIR NET June 2019 MSQ Let $A = (a_{ij})$ be a 3×3 complex matrix. Identify the correct statements.

(A) $\det\left((-1)^{i+j} a_{ij}\right) = \det A$
(B) $\det\left((-1)^{i+j} a_{ij}\right) = -\det A$
(C) $\det\left((\sqrt{-1})^{i+j} a_{ij}\right) = \det A$
(D) $\det\left((\sqrt{-1})^{i+j} a_{ij}\right) = -\det A$